

Solutions: Formula Sheet

Class 12 Chemistry – Complete Revision Formulas

1 Concentration of Solutions

- **Mass Percentage (w/w):**

$$\text{Mass \% of a component} = \frac{\text{Mass of the component in solution}}{\text{Total mass of the solution}} \times 100$$

- **Volume Percentage (V/V):**

$$\text{Volume \% of a component} = \frac{\text{Volume of the component}}{\text{Total volume of solution}} \times 100$$

- **Parts per Million (ppm):**

$$\text{ppm} = \frac{\text{Number of parts of the component}}{\text{Total number of parts of all components}} \times 10^6$$

- **Mole Fraction (x):**

$$x_i = \frac{n_i}{\sum n_i}, \quad x_A = \frac{n_A}{n_A + n_B}, \quad \sum_i x_i = 1$$

- **Molarity (M):**

$$M = \frac{\text{Moles of solute}}{\text{Volume of solution in Litres}} = \frac{w_2 \times 1000}{M_2 \times V \text{ (mL)}}$$

- **Molality (m):**

$$m = \frac{\text{Moles of solute}}{\text{Mass of solvent in kg}} = \frac{w_2 \times 1000}{M_2 \times w_1 \text{ (g)}}$$

2 Gas Solubility & Henry's Law

- **Henry's Law Equation:**

$$p = K_H \cdot x$$

where p = partial pressure of gas in vapour phase, x = mole fraction in solution, and K_H = Henry's law constant.

3 Vapour Pressure & Raoult's Law

- **Partial Vapour Pressure (Raoult's Law for Volatile Liquids):**

$$p_1 = p_1^0 x_1, \quad p_2 = p_2^0 x_2$$

- **Total Vapour Pressure (Dalton's Law of Partial Pressures):**

$$p_{\text{total}} = p_1 + p_2 = p_1^0 x_1 + p_2^0 x_2 = p_1^0 + (p_2^0 - p_1^0) x_2$$

- **Mole Fractions in Vapour Phase (y_i):**

$$p_i = y_i \cdot p_{\text{total}} \implies y_1 = \frac{p_1}{p_{\text{total}}}, \quad y_2 = \frac{p_2}{p_{\text{total}}}$$

- **Thermodynamic Criteria for Ideal Solutions:**

$$\Delta_{\text{mix}} H = 0, \quad \Delta_{\text{mix}} V = 0$$

4 Colligative Properties

- 1. Relative Lowering of Vapour Pressure (RLVP):

$$\frac{p_1^0 - p_1}{p_1^0} = x_2 = \frac{n_2}{n_1 + n_2}$$

For dilute solutions: $\frac{p_1^0 - p_1}{p_1^0} = \frac{n_2}{n_1} = \frac{w_2 \times M_1}{M_2 \times w_1}$

- 2. Elevation of Boiling Point (ΔT_b):

$$\Delta T_b = T_b - T_b^0 = K_b \cdot m$$
$$\Delta T_b = \frac{K_b \times 1000 \times w_2}{M_2 \times w_1} \implies M_2 = \frac{1000 \times w_2 \times K_b}{\Delta T_b \times w_1}$$

- 3. Depression of Freezing Point (ΔT_f):

$$\Delta T_f = T_f^0 - T_f = K_f \cdot m$$
$$\Delta T_f = \frac{K_f \times 1000 \times w_2}{M_2 \times w_1} \implies M_2 = \frac{1000 \times w_2 \times K_f}{\Delta T_f \times w_1}$$

- Thermodynamic Formula for Molal Constants (K_b and K_f):

$$K_b = \frac{R \cdot M_1 \cdot T_b^2}{1000 \cdot \Delta_{\text{vap}}H}, \quad K_f = \frac{R \cdot M_1 \cdot T_f^2}{1000 \cdot \Delta_{\text{fus}}H}$$

- 4. Osmotic Pressure (Π):

$$\Pi = h\rho g = CRT = \left(\frac{n_2}{V}\right) RT$$
$$M_2 = \frac{w_2 RT}{\Pi V}$$

5 Abnormal Molar Masses & van 't Hoff Factor (i)

- van 't Hoff Factor Definitions:

$$i = \frac{\text{Normal molar mass}}{\text{Abnormal (observed) molar mass}}$$
$$i = \frac{\text{Observed colligative property}}{\text{Calculated colligative property}}$$
$$i = \frac{\text{Total moles of particles after association / dissociation}}{\text{Number of moles of particles before association / dissociation}}$$

- Degree of Dissociation (α) & Degree of Association (x):

$$\alpha = \frac{i-1}{n-1}, \quad x = \frac{1-i}{1-\frac{1}{n}}$$

- Modified Colligative Property Equations:

$$\text{RLVP: } \frac{p_1^0 - p_1}{p_1^0} = i \cdot \frac{n_2}{n_1}$$

$$\Delta T_b = i \cdot K_b \cdot m$$

$$\Delta T_f = i \cdot K_f \cdot m$$

$$\Pi = i \cdot CRT = i \cdot \left(\frac{n_2 RT}{V} \right)$$